

# Kaustuv Panthi

737-240-8503 | kaustuvkp@utexas.edu | Austin, TX

## EDUCATION

**The University of Texas at Austin**, Austin, TX

May 2026

*Bachelor of Science, Computer Science*

*Business Minor*

Relevant Coursework: Computer Organization and Architecture, Data Structures, Discrete Math, Probability and Statistics, Machine Learning, Operating Systems, Matrices, Program Correctness and Performance, Multivariable Calculus, Game Development and Design, Object Oriented Programming, Computer Graphics, Linear Algebra

## SKILLS

**Technical/Computer Skills:** *Linux, Microsoft Office, Code Composer, GitHub, OpenAI Prompt Engineering, Cloud Functions, PHPStorm, Visual Studio, VSCode*

**Languages:** *Java, C, C++, Python, React, HTML, PHP, C#, Node.js, MATLAB*

## EXPERIENCE

**Project Seed**, Austin, Texas

September 2024 - Present

*Head Software Developer*

- Architected and maintained cognitive psychology web applications used in academic research studies.
- Implemented front-end systems with React (SPA), integrated Python (NLP pipelines) for language processing tasks, and developed JavaScript game logic consumed in experimental protocols.
- Refactored and stabilized a mission-critical React SPA used for intra-lab communication and data coordination between professors and research assistants.
- Delivered iterative feature updates and defect resolution under weekly sprint reviews with academic stakeholders.

**Paycom**, Dallas, Texas

Summer 2025

*Software Engineering Intern*

- Secured 1st place in company hackathon by engineering a React and C# application for real-time inventory management at client-facing events.
- Developed a React + ASP.NET Core Single Page Application with dynamic UI generation from component libraries and persistent storage in MySQL.
- Designed and deployed backend APIs leveraging the Controller–Service–Repository pattern, implementing CORS authentication and data exchange across PHP and C# services.
- Performed large-scale codebase integration, analyzing and extending a 20+ year monolithic enterprise application while ensuring backward compatibility.

**Paycom**, Dallas, Texas

Summer 2024

*Software Engineering Intern*

- Engineered a React SPA integrated into Paycom's enterprise stack to deliver interactive employee training modules for external client onboarding.
- Implemented Controller-Service-Repository Architecture to create backend API calls with CORS authentication in both PHP and C#

**University of Texas at Austin**, Austin, Texas

May 2021 - August 2021

*EOR Data Analyst Associate*

- Transformed and standardized historical coreflood datasets for integration into modern petroleum engineering simulations.
- Automated Excel workflows for EOR (Enhanced Oil Recovery) models, validating input/output with newly collected experimental data.
- Generated and delivered technical reports to Dr. Kishore Mohanty's research group, supporting graduate-level reservoir simulation studies.

## PROJECTS

**Studia**, Node.js, Python, React

- Designed and implemented the backend architecture for a startup application that leverages OpenAI's GPT models to generate structured lecture content from textbook chapters.
- Integrated LangChain pipelines with embedding-based retrieval to engineer prompts, parse GPT outputs, and deliver context-aware lecture generation.
- Developed RESTful API endpoints in Node.js and optimized GPT query orchestration for consistent, deterministic results.
- Achieved top 10% placement in Y Combinator applications for both Spring 2024 and Winter 2024 cycles.

**Project Assigner**, C#, PHP, React

- Built the backend implementation for an internal application that matched new hires to active projects using resume parsing, skills extraction, and project metadata.
- Applied C# best practices with layered architecture, implementing CORS-secured API endpoints and authentication/authorization workflows.

**Huffman Compression and Decompression Algorithm**, Java

- Engineered a general-purpose compression/decompression utility implementing the Huffman coding algorithm, supporting virtually all file types.
- Managed complex data flows with HashMaps, Binary Trees, and Priority Queues for efficient symbol encoding.
- Optimized I/O operations with bit-level streams, reducing storage footprint and improving read/write efficiency.

**Pintos**, C

- Extended the Pintos educational operating system by implementing:
  - Preemptive threading and scheduling mechanisms.
  - Virtual memory management with page fault handling.
  - A multi-level file system supporting persistence and metadata indexing.
  - Syscall interfaces enabling seamless user program execution.
- Validated system robustness under concurrency stress tests and I/O workloads.

**Memory Manager**, C

- Developed a custom dynamic memory allocator replicating malloc() and free() semantics.
- Implemented block splitting, coalescing, and free-list management for efficient allocation.
- Tuned allocator design to sustain 6000+ operations per millisecond while maintaining >80% space utilization efficiency.
- Demonstrated deep expertise in pointer arithmetic, heap management, and memory fragmentation control.

**Laser Tag Tag**, Unreal Engine 5

- Collaborated with a team of four developers to design and build a multiplayer laser tag game in Unreal Engine 5.
- Implemented replicated gameplay systems with full remote play support, ensuring deterministic state synchronization across clients.
- Integrated Steamworks API for online hosting, matchmaking, and lobby management.
- Leveraged UE5's core systems (Blueprints, networking, and physics simulation) to deliver real-time multiplayer gameplay.